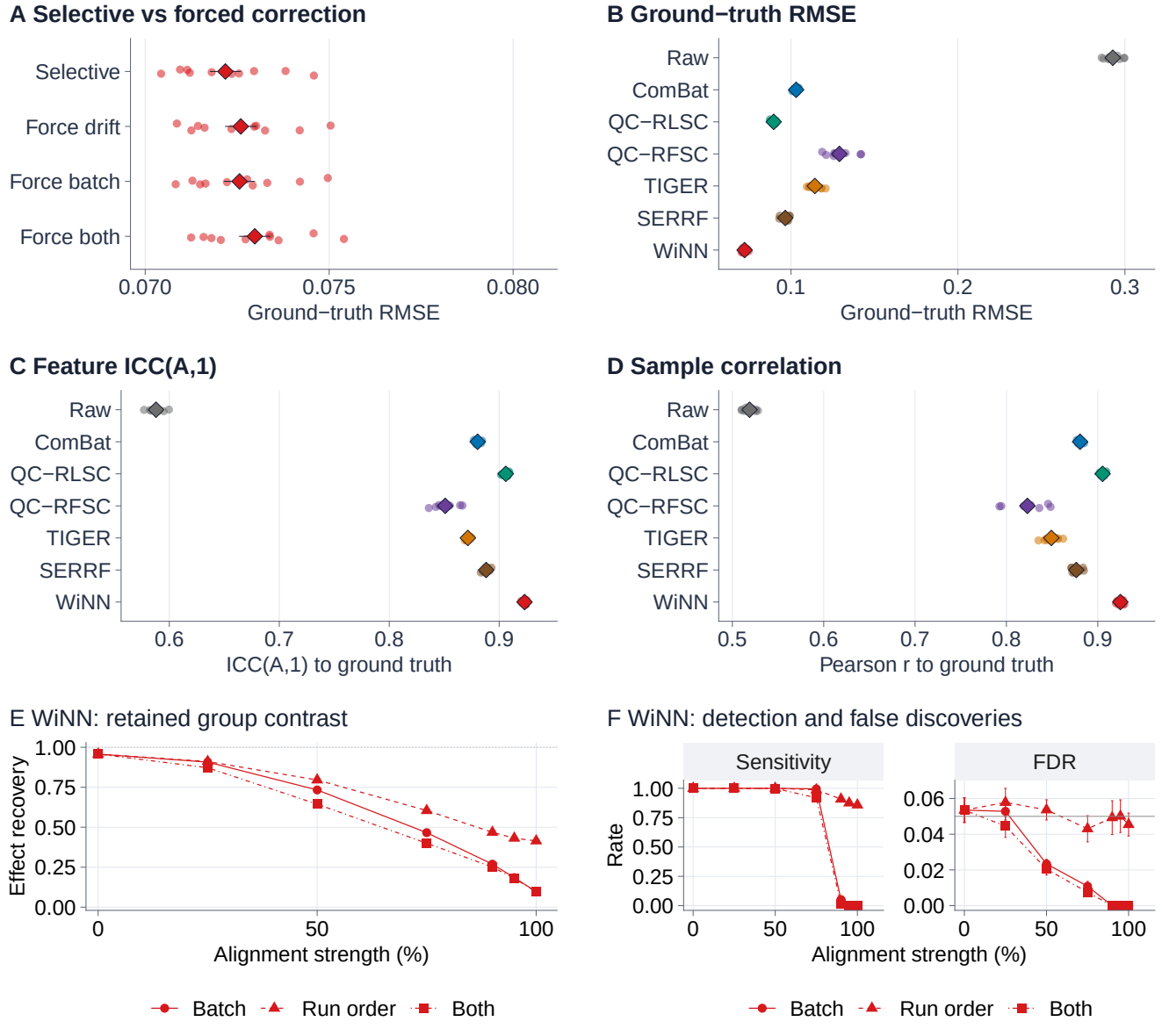

A Selective vs forced correction
Selective
Force drift
Force batch
Force both
0.070
0.075
0.080
Ground−truth RMSE
B Ground−truth RMSE
Raw
ComBat
QC−RLSC
QC−RFSC
TIGER
SERRF
WiNN
0.1
0.2
0.3
Ground−truth RMSE
C Feature ICC(A,1)
0.6
0.7
0.8
0.9
ICC(A,1) to ground truth
D Sample correlation
0.5
0.6
0.7
0.8
0.9
Pearson r to ground truth
E WiNN: retained group contrast
Effect recovery
1.00
0.75
0.50
0.25
0.00
0
50
100
Alignment strength (%)
Batch
Run order
Both
F WiNN: detection and false discoveries
Sensitivity
FDR
Rate
1.00
0.75
0.50
0.25
0.00
0.06
0.04
0.02
0.00
0
50
100
Alignment strength (%)
Batch
Run order
Both